



Maxime Dherbecourt

PhD Student in Physics

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PROFILE

I'm a PhD student based at the IPCMS laboratory in Strasbourg (France), working on strong light-matter coupling and its effects on physical phenomena such as Raman scattering and transport properties. I enjoy making science accessible through interactive explorations and visual explanations.

Research interests: Light-Matter Interactions · Condensed Matter · Open Quantum Systems · Numerical Methods

EDUCATION

PhD in Physics – Strong Light-Matter Coupling Institut de Physique et Chimie des Matériaux de Strasbourg (IPCMS) · France Advisors: Guillaume Weick and David Hagenmüller.	2024 – Now
Master Degree – International Center for Fundamental Physics (ICFP) Ecole Normale Supérieure de Paris · France Condensed Matter track. Graduated with honors.	2023 – 2024
Magistère de Physique Fondamentale Université de Strasbourg · France	2021 – 2024
Bachelor Degree – Advanced Mathematics and Physics Université de Strasbourg · France	2019 – 2022

TEACHING

Electromagnetism 1 Lecturer · Université de Strasbourg – General Science License	2024 – 2026
French Physicists' Tournament Assistant · Université de Strasbourg – Master of Physics	2024 – 2026

INITIATIVES

French Physicists' Tournament Organizer · Young Researcher Network – Société Française de Physique	2025 – Now
Winner of a science popularization contest Prize Winner · Summer of Math Exposition – "The Science of Complexity"	2025

Scientific Game Jam on my PhD project 2025
Participant · Strasbourg – Created game "Molec' Adventure"

PRESENTATIONS

Raman Scattering in the Vibrational Strong Coupling Regime 2026
Poster · Summer School – Open Quantum Systems, Les Houches, France

Raman Scattering in the Vibrational Strong Coupling Regime 2026
Poster · QMat Days, Strasbourg, France

Spontaneous Raman scattering under Vibrational Strong Coupling 2026
Contributed Talk · European Materials Research Society Conference, Strasbourg, France

Spontaneous Raman scattering under Vibrational Strong Coupling 2026
Invited Talk · Namasute Workshop, Odense, Denmark

Spontaneous Raman scattering under Vibrational Strong Coupling : The critical role of polariton spatial mode coherence 2025
Invited Talk · $\hbar$ -Meeting, Strasbourg, France

PUBLICATIONS

Spontaneous Raman Scattering under Vibrational Strong Coupling: The Critical Role of Polariton Spatial Mode Coherence 2026
M. **Dherbécourt**, J. Bellessa, C. Symonds, G. Weick and D. Hagenmüller
arXiv preprint, arXiv:2511.01723 · doi.org/10.48550/arXiv.2511.01723

Symmetry-controlled ultrastrong phonon–photon coupling in a terahertz cavity 2026
D. Kim, M. **Dherbécourt**, et al.
The Journal of Chemical Physics, 164, 104201 · doi.org/10.1063/5.0313120

PERSONAL PROJECTS

Archive Explorer 2026
A website to keep track of the games you played, the books you read, and the movies and series you watched. It allows you to create a personal archive of your experiences and share them with others.
Web Development · Personal Archive · Next.js · TypeScript · Tailwind CSS · archive-explorer.mdherbecourt.dev

WaterDropEngine Terrain Generator 2026
A procedural terrain generator built with WaterDropEngine, producing realistic 3D landscapes using noise-based algorithms and GPU-accelerated mesh generation.
Rust · Procedural Generation · Terrain · Vulkan · github.com/explorablescience/WaterDropTerrainGenerator

Portfolio Website 2025
The source code for this portfolio website, built with Next.js, TypeScript, Tailwind CSS and Vercel. It features a clean and modern design, showcasing my projects and skills.
Next.js · TypeScript · Tailwind CSS · Vercel · React · github.com/xam4lor/Website

WaterDropEngine 2023
A high-performance, ECS driven and Vulkan based game engine written in Rust. Designed for flexibility and speed, with a focus on 3D game development.
Rust · Game Engine · Vulkan · WebGPU · github.com/explorablescience/WaterDropEngine

Explorable Science 2019
Explorable visual articles and interactive simulations on various scientific topics such as phase transitions, general relativity, quantum mechanics and more.

Physics Simulation Engine

2017

A high-level JavaScript library for creating interactive physics simulations in the browser built on top of the canvas API.

JavaScript · Physics Simulation · Educational · Interactive Science · github.com/explorable.science/PhysicsSimulationEngine

SKILLS

Programming Languages: Julia, Rust, JavaScript/TypeScript, HTML5/CSS3, C++, Python3, Java

Programming Libraries & Frameworks: Vulkan / WebGPU / OpenGL, PyTorch / NumPy, Node.js / React.js / Three.js / React Three Fiber / Next.js, Plotly.js / p5.js / Processing, CUDA

Scientific tools: Mathematica, Maple

Other: LaTeX, Git, Docker

LANGUAGES

French: Native · **English:** Fluent (C1) · **German:** Intermediate (B1)